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# Quentin Rebergue - C++ Systems Engineer

## Profile

Looking for a full-time position in software engineering with a focus on low-latency C++ systems and infrastructure. MSc in Computer Engineering (EPITA - GISTRE) in Systems / Embedded. Fluent in French and English (TOEIC 915, C1). Interests: watchmaking, music, hiking/running, cooking.

## Work Experience

- **Software Engineer - Platform team, ARISTA**, Dublin 02/2026 – Now

  Implemented non-zero reset value management for ASIC Physical Table registers across the chip-agnostic Strata infrastructure (SDK parsing, code generation, DMA) in python and C++. Won 1st place at Arista's Hackathon by building an AI agent optimization tool (70% token reduction on internal codebases). Approved for production development.

- **Head of Teaching Assistants - YAKA** / **ACU Program, EPITA**, Kremlin-Bicetre 09/2024 – 02/2026

  Led the national team of 67 TAs and supervised 5 intensive coding bootcamps (Java, C++, JS, C/Bash, SQL) and 7 large-scale projects (compiler, HTTP server in C, malloc, POSIX shell) for 700+ 3rd-year students across 5 campuses. Implemented 1-on-1 code review mentorship model, improving TA-to-student ratio from 1:30 to 1:20.

- **Freelance Software Engineer, Various Clients**, Paris 09/2021 – 03/2024

  Built and managed full-stack information systems for 10+ clients. Led B2B survey platform (200k euros, 2 devs) as first technical lead. Architected PIM + CRM + ERP ecosystem. Owned end-to-end IS at Aussitôt Bon.

## School Projects

- **GPU-Accelerated Video Analysis System (IRGPU)** – Built a real-time motion detection pipeline in C++/CUDA (reservoir sampling, morphological ops, hysteresis thresholding). Achieved 20x CPU speedup through iterative CUDA optimization profiled with Nsight Compute. Team of 4.
- **Tiger Compiler** – Built compiler frontend in C++: lexing, parsing, binding, type checking, desugaring and LLVM IR. Team of 4.
- **My Docker** – Built a lightweight container runtime in C, using Linux namespaces, pivot_root, capabilities dropping, seccomp syscall filtering, and cgroups. Pulls and mounts a root filesystem from a container registry.
- **RFID Driver** – Developed a Linux kernel character device driver for the MFRC522 RFID reader with file_operations callbacks, SPI regmap communication, and a FIFO command interface. Tested via QEMU emulation.

## Education

- **MSc Computer Engineering, Low-level & Systems (GISTRE)**, EPITA 2020 – 2026
- **Preparatory Class (PTSI) in Engineering Science**, Lycée Jules Ferry Versailles 2019 – 2020

## Technical Expertise

- **Programming & Languages:** ASM, C, C++, OOP, Python, Rust, SQL, VHDL.
- **Other** / **DevOps:** GDB, valgrind, perf, profiling, optimisation, strace, system analysis, testing, benchmarking.
- **Systems:** ASIC interfacing, Linux, OS, TCP/IP, cache coherence, kernel, low-latency, memory architecture, multithreading, networking, profiling, real-time, shared memory, virtual memory.